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# A high-speed MCML charge pump design at 10 GHz frequency in 45 nm CMOS technology for PLL application

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**Abstract**
In this paper, a new high speed two-stage charge pump is designed for phase-locked loop (PLL) application. In the proposed circuit, switch-based charge pump acts as the primary charge pump for glitch-free output, in addition to that MOS current mode logic (MCML) based faster current driving charge pump acts as the secondary charge pump. It will used to achieve the PLL locking condition quickly. MCML circuits minimize delay and perform the fast operation, hence it can be used in high frequency applications. The proposed circuit achieves very low power of 13.19 µW with a minimum delay of 16.71 ps at 45 nm CMOS technology with a 1 V power supply in different process corners. The output noise as very low as − 232.7 dB and phase noise as − 247.2 dBc/Hz at 10 GHz frequency. The swing voltage ranges from 0 to 980 mV. Monte-Carlo simulations with 200 samples are analysed to verify the results. Finally, process voltage and temperature (PVT) analysis are performed to validate the stability of the proposed design. The simulated results shows that the proposed circuit is more stable for high-frequency PLL applications and is highly tolerant with PVT variations.

**Keywords** Charge pump (CP) · Phase locked loop (PLL) · Switch-based charge pump · Faster current driving charge pump (FCD-CP) · MOS current mode logic (MCML)

## 1 Introduction

Many modern microprocessors and telecommunication systems now using phase-locked loops (PLLs) in clock pulse generators (CPG) and frequency synthesizers (FS). Charge pump PLL are the most common type of PLL [1–3]. Analog and digital are the two fundamental approaches in designing PLLs for multi-GHz applications. Analog PLLs typically consume more power compared to their digital PLLs, known as DPLLs. As an alternative, DPLLs have greater complexity and are often restricted above 5 GHz frequencies. Recently, analog type of PLLs have slowly added in the digital building blocks such as oscillators and phase detectors which consumes less area and power [4–6]. To achieve high speed in the mixed signal environment MOS current mode logic (MCML) circuits are used [7, 8].

Figure 1 depicts a conventional PLL block diagram. The charge pump is necessary to maintain the stability in the PLL design [9, 10]. The loop filter produces the control voltage ($V_{cp}$) that regulates the frequency of the oscillator. When the dynamic range of $V_{cp}$ reduces, using cascade transistors becomes preferable to increase the output impedance [11]. To simplify clock and data recovery (CDR) systems, it is essential to create a Phase Frequency Detector (PFD) with an extensive input range that achieves improvements in both speed and power efficiency. Additionally, designing the CDR encounters challenges in maintaining a consistent matching current for charging and discharging [12, 13]. Most CDR applications choose PFDs instead of PD's because they monitor both frequency and phase at the same time [14].

An op-amp based charge pump is introduced with current mirror circuit which produces wide output range of voltage and also fasten the PLL settling time by increasing the charge current [15]. A charge pump with high output impedance is designed using a low-voltage current mirror setup. This configuration ensures excellent current matching while minimizing transient glitches [16]. A PVT tolerant

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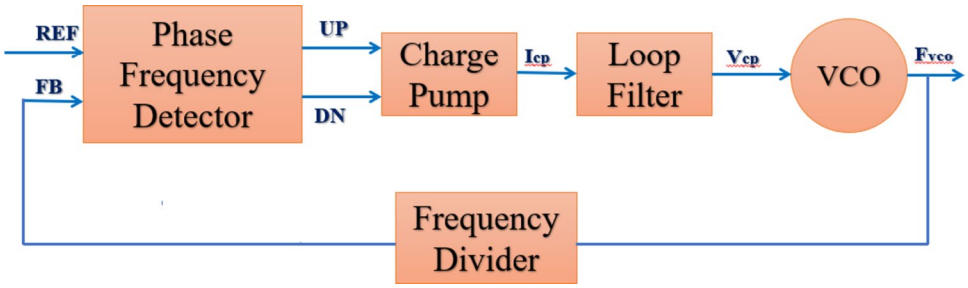



**Fig. 1** Block diagram of Phase Locked Loop

hybrid current starved ring VCO exhibits better power and phase management with the bulk driven technique using MCML [17]. A switch based charge pump circuit provides better results but minimizing delay and phase noise is a challenging one [18]. Another technique involves employing dynamic current matching and transmission gate circuits alongside the cascode current mirror approach to implement low-noise PLLs [19, 20].

The charge pump circuit design aligns with the gate's switching architecture. Negative feedback circuits are integrated to counteract channel length modulation [21]. In high-speed locking PLL circuits, the Miller operational amplifier is preferred for its ability to enhance operating frequency and gain. It serves as a compensation network, ensuring the stability of the circuit [22]. The charge pump circuit responses based on the PFD output signals. Having a reset path in a typical PFD not only causes disturbances in the lock condition and also limits the maximum speed. The typical PFD uses a lot of power at higher frequencies. Various designs were designed and executed in order to solve these drawbacks. When the input frequency difference between the two inputs is quite large, the pre-charge PFD fails to find some reference signal. However, adding delays in feedback which increases the dead zone [23]. To eliminate the dead zone and generate a low-glitch output, a tri-state PFD circuit was designed. However, it comes at the cost of increased area consumption and is effective primarily in a low-frequency range [24].

### 1.1 PFD charge pump overview

The reset-free phase frequency detector ensures a dead zone-free condition without requiring a reset. This circuit consumes significantly less area when compared to PFD circuits that incorporate reset mechanisms [25]. The proposed circuit uses this reset free PFD design to get the UP and DOWN outputs as inputs to the charge pump circuit. T and Δt are two input signals for time period and time difference respectively. Hence the phase difference ($\Delta_{\varphi}$) is,

$$\Delta\varphi = \frac{\Delta t}{T} \tag{1}$$

When the input phase difference is zero, the PFD output voltage with the supply voltage $V_{DD}$ is,

$$\text{VPFD} = \frac{V_{DD} - V_{dif}}{4\pi} X\Delta\varphi \tag{2}$$

The phase frequency detector design consists of pmos transistors $MP_1$ to $MP_8$ and nmos transistors $MN_1$ to $MN_8$. This reset free PFD works in 3 cases.

*Case 1*: Initially, $f_{ref}$ and $f_{vco}$ must be 0 to turn pmos transistors becomes ON and nmos transistors becomes OFF which produces 1 at the output $u_1$, this makes UP and DN output becomes 0.

*Case 2*: When $f_{ref}$ leads $f_{vco}$ signal, UP signal becomes high and DN signal maintains 0.

*Case 3*: When $f_{ref}$ lags $f_{vco}$ signal, DN signal becomes high and UP signal maintains 0.

The proposed charge pump circuit is simulated with the UP and DN results obtained from this dead zone free PFD design which is simulated with 45 nm CMOS technology in Cadence virtuoso tool.

In this paper, a new faster current driving charge pump (FCD-CP) circuit is proposed. This charge pump works in 3 conditions: (1) Up = Down = 0 called idle condition, (2) Up = 1, Down = 0 called charging condition and (3) Up = 0, Down = 1 called discharging condition.

The average power of the proposed circuit is 13.19 µW with a minimum delay of 16.71 ps at 45 nm CMOS technology with 1 V power supply in different process corners. The output noise is − 232.7 dB and phase noise − 247.2 dBc/Hz. The swing voltage of the proposed charge pump is 0 to 980 mV.

The percentage of performance improvement interms of Average Power, Propogation Delay, Noise, Swing Voltage and Power Delay Product is 39.55%, 76.57%, 8.61%, 98% and 85.8% respectively. The results are analysed with Monte Carlo simulations and parametric analysis is also performed with different temperature and voltages to verify the stability of the proposed charge pump.



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The remaining section is structured as: Sect. 2 describes the existing charge pump circuit with the mathematical equations based on the literature, Sect. 3 elaborates the proposed circuit with working principle, Sect. 4 analyses the simulation results, performance comparison with different charge pump circuits and Sect. 5 discusses the conclusion.

## 2 Related work

The switch based charge pump circuit consists of 4 transistors ($S_1$, $S_2$, $S_3$ and $S_4$), an interconnect capacitor $C_i$, output voltage $V_{op}$ and supply voltage $V_{DD}$ [18] shown in Fig. 2. Depends upon the PFD output, the charge pump works in the following conditions. Initially when UP = DN = 0, then PFD output becomes zero. During this condition, $S_1$ and $S_2$ becomes ON and $S_3$ and $S_4$ switches becomes OFF hence $V_{op}$ remains in the previous state.

When UP = 1 and DN = 0, the charge pump works in the charge mode and capacitor $C_L$ charged Via $S_1$-$C_i$-$S_4$. The voltage $V_{op}$ can be expressed as [18],

$$V_{op} = V_{op(0)} + \frac{C_i}{C_i + C_L}(V_{DD} - V_{op(0)})(1 - e^{\frac{-t}{\tau_{UP}}}) \tag{3}$$

$\tau_{UP}$ equals to, $\tau_{UP} = R_{UP}C_V$ (4)

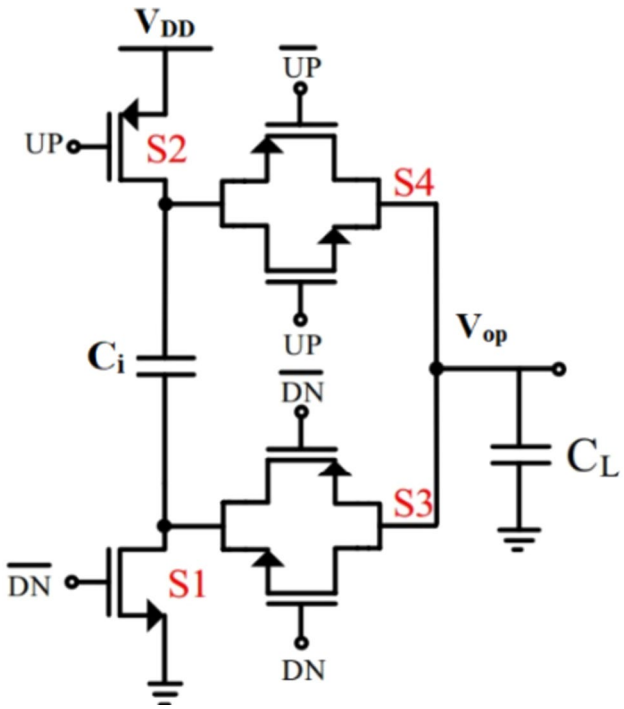



**Fig. 2** Switch-based charge pump circuit [18]

$$V_{op} \approx V_{op(0)} + \frac{C_i}{C_i + C_L}V_{DD}\left(1 - \left(1 - \frac{t}{\tau_{UP}}\right)\right) \tag{5}$$

$$V_{op} \approx V_{op(0)} + \frac{C_i}{C_i + C_L}V_{DD}\left(\frac{t}{\tau_{UP}}\right) \tag{6}$$

Equation 6, proves that, the output voltage can charge to produce the VCO output frequency.

When UP = 0 and DN = 1, the charge pump works in the discharge mode and capacitor $C_L$ discharged Via $S_2$-$C_i$-$S_3$. The voltage $V_{op}$ can be expressed as [18],

$$V_{op} = V_{op(0)} + \left(1 - \frac{C_i}{C_i + C_L}\right)(1 - e^{\frac{-t}{\tau_{DN}}}) \tag{7}$$

$\tau_{DN}$ equals to, $\tau_{DN} = R_{DN}C_i$ (8)

$R_{DN}$ is the summation of channel resistance. Near to the locking condition ($t << \tau_{DN}$) Eq. 7 can be written as,

$$V_{op} \approx V_{op(0)} - \frac{C_i}{C_i + C_L}V_{op(0)}\left(1 - \left(1 - \frac{t}{\tau_{DN}}\right)\right) \tag{9}$$

$$V_{op} \approx V_{op(0)}\left(1 - \frac{C_i}{C_i + C_L}\left(\frac{t}{\tau_{DN}}\right)\right) \tag{10}$$

## 3 Proposed faster current driving charge pump (FCD-CP)

The proposed FCD-CP consists of,

(1) Primary Charge Pump
- Switch-based charge pump ($S_1$, $S_2$, $S_3$ and $S_4$ switches along with capacitor $C_i$).

(2) Secondary Charge Pump
- MCML based Faster Current Driving Charge pump (FCD-CP) ($M_1$, $M_2$, $M_3$, $M_4$, $M_5$, $S_5$ and $S_6$).

Figure 3 shows the circuit of proposed charge pump. It can achieve faster current driving condition for PLL application. It mitigates phase noise and propagation delay. When the PFD is significant the charge pump offers charging or discharging current based on the outputs from the PFD (UP/Down). When the reference signal (fref) leads VCO signal (fvco) the PFD output becomes UP = 1 and Down = 0. When the VCO signal (fvco) leads reference signal (fref) then the PFD output becomes UP = 0 and Down = 1.

The switch-based charge pump produces glitch free output signal. It consists of $S_1$ and $S_2$ transistors as switches and interconnect capacitor $C_i$ along with transmission gate



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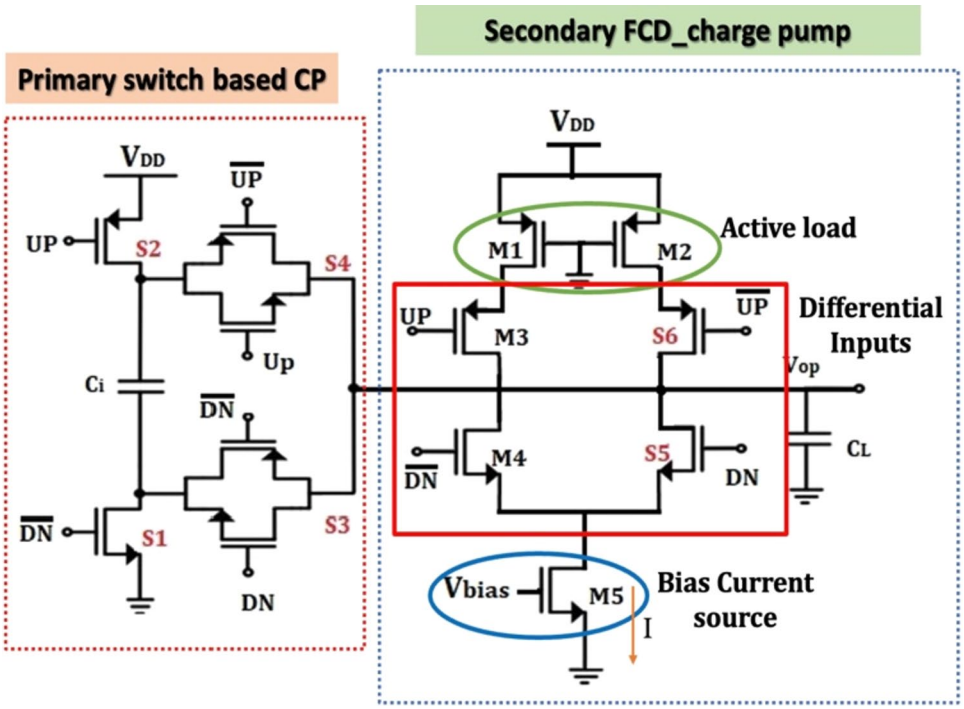



**Fig. 3** Proposed Faster current driving charge pump (FCD-CP) circuit

arrangement as $S_3$ and $S_4$ switches to transfer the charges. The secondary charge pump is used to achieve the faster current driving condition. It consists of $M_1$ and $M_2$ transistor as pmos load and $M_5$ transistor acts as current source. $S_5$ and $S_6$ switches provides necessary charging and discharging currents to the capacitor $C_L$. The differential inputs of up and down are applied to the transistors $M_3$, $M_4$, $S_5$ and $S_6$ because of MOS current mode logic arrangement. Both primary and secondary charge pumps having individual charging and discharging paths which accelerates the locking process quickly.

The proposed charge pump operates in 3 conditions:

(1) Idle condition
(2) Charging condition
(3) Discharging condition

## 3.1 Operation of the proposed charge pump

### 3.1.1 Idle condition "UP = DN = 0"

When both Up and Down signals from PFD are 0 called idle condition, then $S_1$, $S_2$, $S_3$ and $S_4$ switches in the primary charge pump becomes OFF and $M_3$, $M_4$ becomes ON but $S_5$ and $S_6$ becomes OFF hence no transition takes place $C_L$ remains in the previous state.

### 3.1.2 Charging condition "UP = 1 and DN = 0"

Figure 4 shows the operation and transient response of charging condition. At this condition, $S_2$, $S_4$ and $S_6$ switches becomes ON and $S_1$, $S_3$ and $S_5$ becomes OFF. Initially when the charge pump receives input, $S_2$ PMOS transistor becomes ON and $S_1$ NMOS transistor becomes OFF. The transmission gate switches $S_4$ becomes ON and $S_3$ becomes OFF. It transmits the charge from $S_2$ to the output node via $S_4$. At the same time, $S_6$ PMOS transistor becomes ON and $S_5$ NMOS transistor becomes OFF. The transistor $S_6$ is connected to $V_{DD}$ via the PMOS load, when the input UP = 1and $\overline{UP}$ =0, $S_6$ PMOS becomes ON. This transfers the charge to the output node. Hence the load capacitor is charged to $V_{DD}$. The capacitor charging waveform is shown in Fig. 6a.

To achieve fast locking condition, and avoid transient response delay both charge pumps are processed simultaneously. Because of the load capacitor $C_L$ is in charging condition the output voltage $V_{op}$ swings up to a certain level of $V_{DD}$. The simulated results shows that proposed charge pump swings upto 980 mV for 1 V supply.

Based on the literature, the required gate potential for the transistor $M_6$ is,

$$V_{G6} \leq V_{dd} - |V_{thp}| \tag{11}$$



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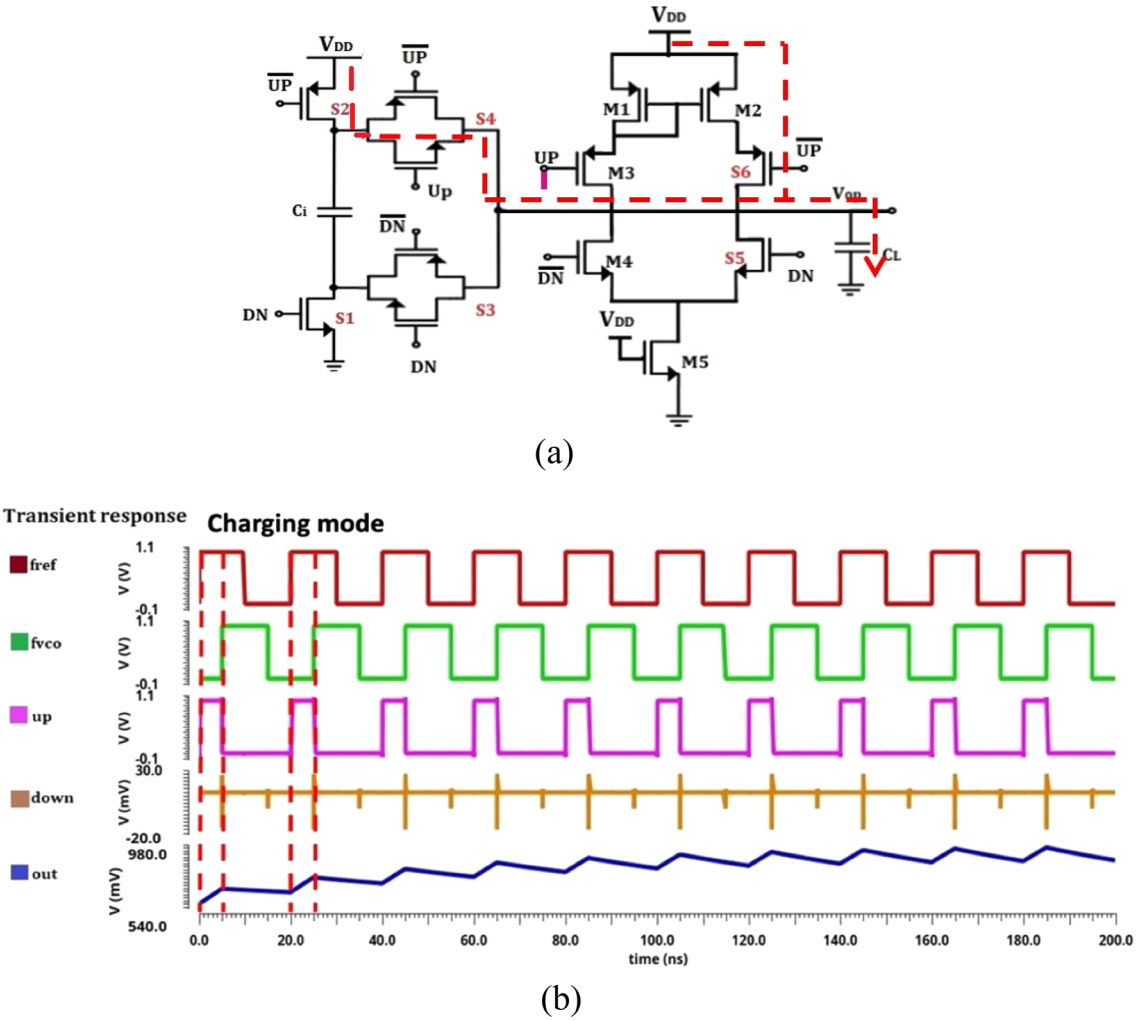



**Fig. 4** Proposed charge pump in the charging condition **a** Operation of the circuit **b** Transient Response of the proposed charge pump

The drain current flows across the transistor $M_6$ is

$$I_{FCD_up} = \frac{1}{2}\mu_P C_{ox}\left(\frac{W}{L}\right)_{M6}(V_{dd} - V_{thp})^2 \tag{12}$$

$I_{FCD_up}$ is the faster current driving charge pump current which is used to drive the proposed circuit quickly during up charging.

The output potential at the charge pump is,

$$V_{op} = V_{op(0)} + \frac{C_i}{C_i + C_L}\left(V_{dd} - V_{op(0)}\right) + \frac{I_{FCD_CP}}{C_i + C_L} \tag{13}$$

#### 3.1.3 Dis-charging condition "UP = 0 and DN = 1"

Figure 5 shows the operation and transient response of discharging condition. At this condition, $S_1$, $S_3$ and $S_5$ switches becomes ON and $S_2$, $S_4$ and $S_6$ becomes OFF. Initially when the charge pump receives input, $S_1$ NMOS transistor becomes ON and $S_2$ PMOS transistor becomes OFF. The transmission gate switches $S_3$ becomes ON and $S_4$ becomes OFF. It transmits the charge from $S_1$ to the output node. At the same time, the $S_5$ NMOS transistor becomes ON and $S_6$ PMOS becomes OFF. The transistor $S_5$ is connected to GND via the NMOS current source,



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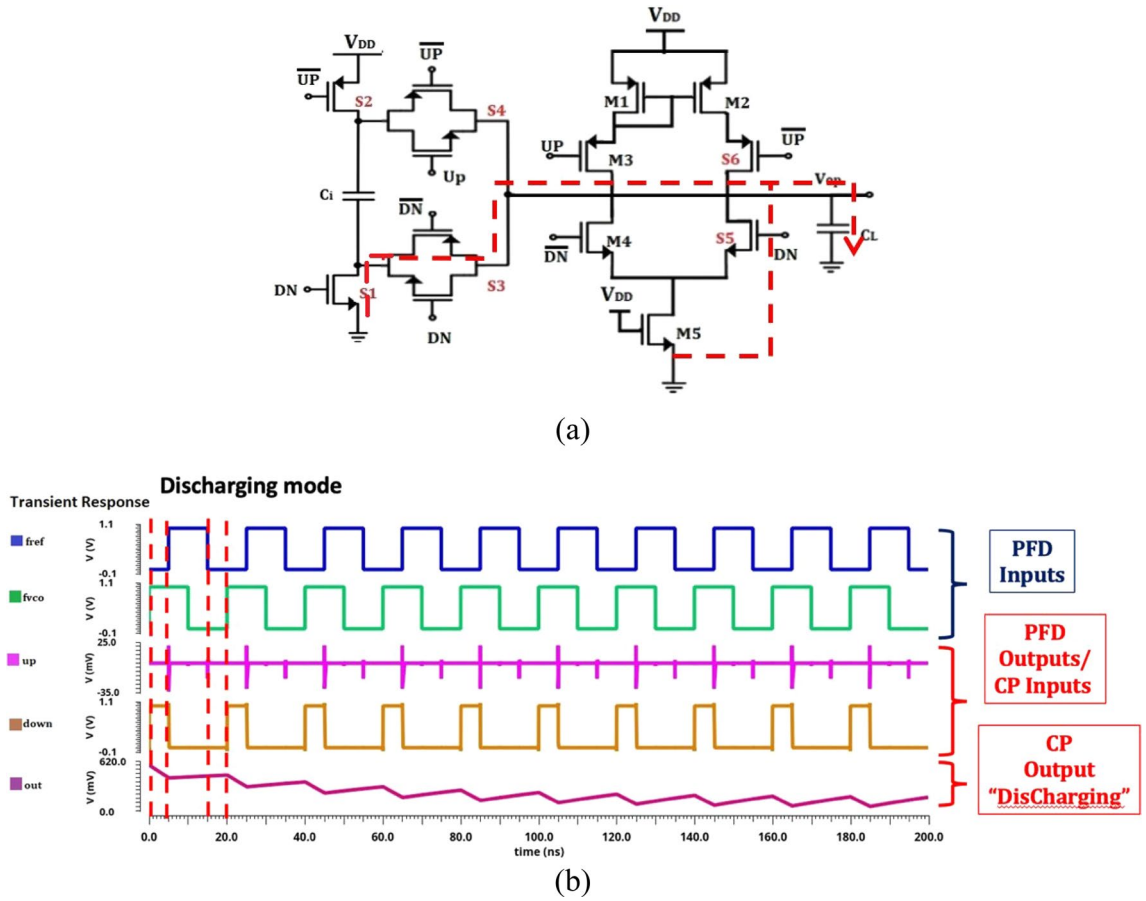



**Fig. 5** Proposed charge pump in the discharging condition **a** Operation of the circuit **b** Transient Response of the proposed charge pump

when the input DN = 1 and $\overline{DN}$ =0, $S_5$ NMOS becomes ON. This transfers the charge from $M_5$ to the output node. Hence the load capacitor discharged to GND. The capacitor discharging waveform is shown in Fig. 6b. Because of the load capacitor $C_L$ is in discharging condition the output voltage $V_{op}$ swings down to a certain level. The simulated results shows that proposed charge pump swings down to a maximum of 0 V.

Based on the literature, the required gate potential for the transistor $M_5$ is,

$$V_{G5} \geq V_{thn} \tag{14}$$

The drain current flows across the transistor $M_5$ is

$$I_{FCD_down} = \frac{1}{2}\mu_n C_{ox}\left(\frac{W}{L}\right)_{M5}(V_{dd} - V_{thn})^2 \tag{15}$$

$I_{FCD_down}$ is the faster current driving charge pump current during down charging.

The output potential at the charge pump is,

$$V_{op} = V_{op(0)}\frac{C_L}{C_i + C_L} - \frac{I_{FCD_down}}{C_i + C_L} \tag{16}$$

### 3.1.4 Sizing of the transistors

In the proposed charge pump design, the transistors are sized based on the analysis performed with the power, delay and noise for high frequency applications of PLL. Optimal transistor sizing involves, finding the balance between speed, power and noise becomes a challenging one. Larger transistors can offer faster switching but might consume more power during the state changes. Smaller transistors generally exhibit lower parasitic capacitance, thereby reducing the impact of thermal noise, a major contributor to output noise and phase noise. So based on the simulation results obtained from the trial and error method of the proposed charge pump, the transistor sizing is fixed with the length of 45 nm and with the optimized width for better power, delay and noise management. Figure 7a–c illustrates



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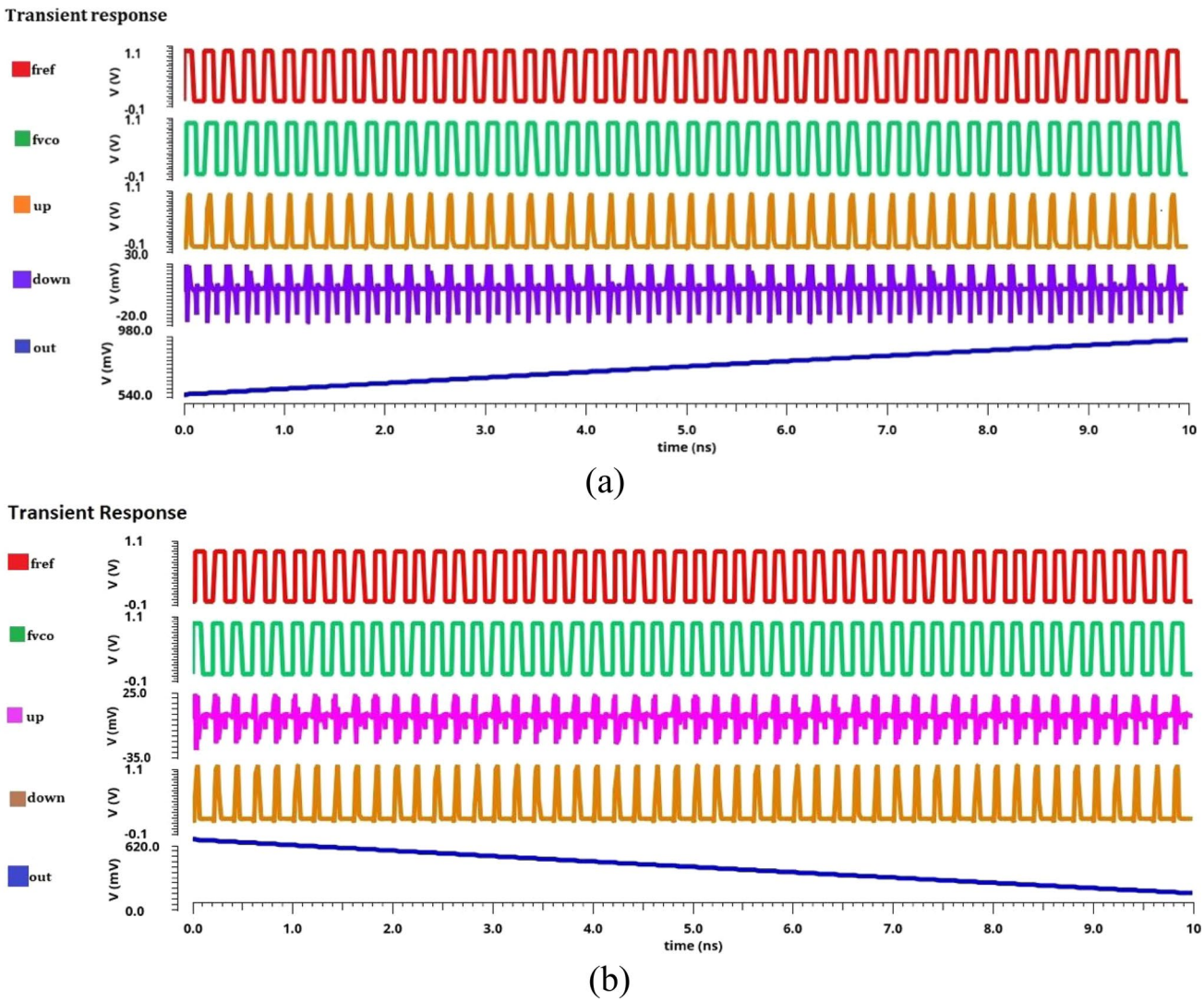



**Fig. 6** Transient response at 10 GHz during **a** Charging condition **b** Discharging Condition

changes in power, delay and noise by increasing the width of transistors ($S_1$, $S_2$, $M_1$ and $M_2$, $M_1$ and $S_6$, $M_4$ and $S_5$ and $M_5$).

For the transistors $S_1$ and $M_5$, as width increases the power and phase noise reduces but delay increases. For the transistors $M_1$ and $M_2$, as width increases delay decreases but power reduces till 200 nm and after it increases and noise slightly increases. For the transistors $M_3$ and $S_6$, as width increases, delay is nearly constant, phase noise slightly reduces but power increases as width hence width is selected as minimum. Similarly, the analysis involves adjusting the width of individual transistors to examine the variations in power, delay and noise and finally the optimized transistor width is fixed for each transistor represented in Table 1.

## 4 Results and discussion

The existing and the proposed charge pump circuits are verified with the simulation environment represented in Table 2.

### 4.1 Analysis of power, propagation delay and power delay product

MCML technique does not consumes much power at higher frequencies. It can achieve faster switching speeds because of the current steering techniques. Faster switching can reduce the active time and consequently decrease power consumption. Table 3 analyses the results of power, PDP, delay and noise in different process corners. The simulation results of existing and proposed charge pump are achieved using cadence virtuoso with a voltage of 1 V



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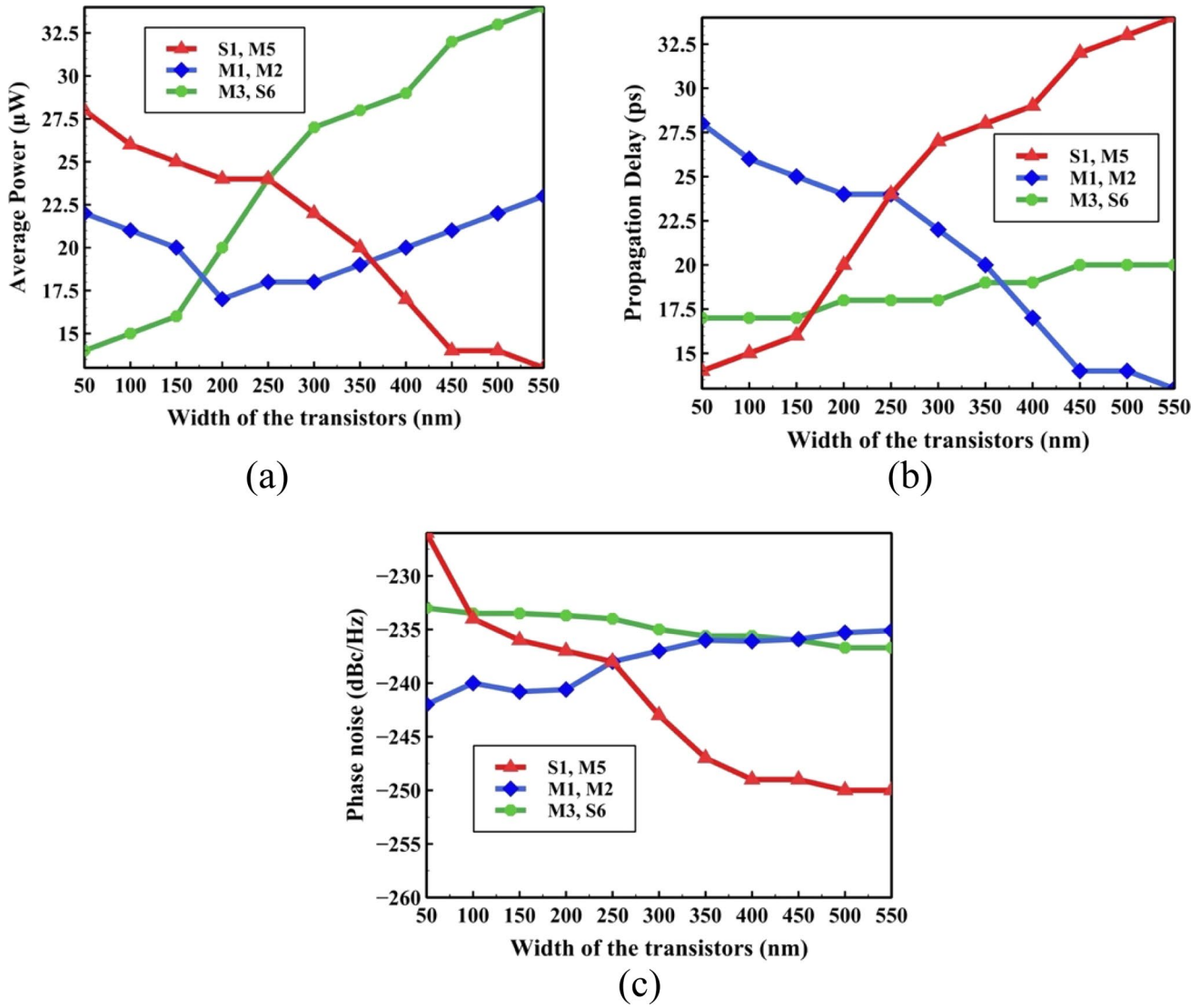



**Fig. 7** Analysis of transistors width for **a** Average power **b** Propagation delay and **c** Phase noise

**Table 1** Sizing of transistors for the proposed charge pump

| Primary charge pump | | Secondary charge pump | |
|---|---|---|---|
| Transistors | Width of the transistors (nm) | Transistors | Width of the transistors (nm) |
| $S_1$ | 360 | $M_1$ and $M_2$ | 190 |
| $S_2$ | 170 | $M_3$ and $S_6$ | 180 |
| $S_3$, $S_4$ (pmos) | 130 | $M_4$ and $S_5$ | 270 |
| $S_3$, $S_4$ (nmos) | 280 | $M_5$ | 480 |

**Table 2** Simulation environment

| CMOS technology | 45 nm |
|---|---|
| Supply voltage | 0.8 V, 1 V, 1.2 V |
| Process corner | TT, FF, SS, FS, SF |
| Capacitance | 10 pF |
| Temperature | – 90 °C, – 40 °C, 0 °C, 27 °C, 90 °C |
| Simulation environment | Cadence Virtuoso |

and temperature 27 °C at 45 nm CMOS technology. The average power of the proposed circuit is 13.19 µW with a minimum delay of 16.71 ps and a power delay product of 0.225fJ.

## 4.2 Monte Carlo analysis

Figure 8 shows the histogram plot of Monte-Carlo simulations. It is clear that, the mean value of the power is



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**Table 3** Corner analysis of existing and proposed charge pump at 45 nm CMOS technology

| Different techniques of charge pump | Different process corner | Parameters of the charge pump | | | | |
|---|---|---|---|---|---|---|
| | | Power (μW) | Delay (ps) | Power delay product (fJ) | Output noise (dB) | Phase noise (dBc/Hz) |
| Existing charge pump [18] | TT | 21.82 | 71.34 | 1.56 | − 214.8 | − 227.6 |
| | FF | 30.56 | 63.78 | 1.95 | − 212.3 | − 224.8 |
| | SS | 13.1 | 77.19 | 1.01 | − 218.5 | − 222.9 |
| | FS | 18.27 | 69.49 | 1.27 | − 215.7 | − 228.5 |
| | SF | 22.38 | 75.61 | 1.69 | − 211.1 | − 220.3 |
| Proposed charge pump | TT | 13.19 | 16.71 | 0.221 | − 232.7 | − 247.2 |
| | FF | 18.44 | 11.54 | 0.213 | − 234.1 | − 245.3 |
| | SS | 6.01 | 21.39 | 0.128 | − 232.4 | − 243.7 |
| | FS | 8.22 | 13.76 | 0.113 | − 232.7 | − 247.8 |
| | SF | 15.4 | 18.45 | 0.284 | − 232.8 | − 241.4 |

13.164 μW which is closer to the TT (Typical) results, that represents the stability of the proposed design. The delay value of the histogram plot represents 16.65 ps, power delay product of 0.224 fJ, output noise of − 231.089 dB and phase noise of − 247.524 dBc/Hz.

### 4.3 Noise analysis

Figure 9 shows the output noise and phase noise response of the proposed charge pump. The phase noise analysis is performed using the periodic steady state with beat frequency of 10 GHz [26]. The value of output noise and phase noise are − 232.7 dB and − 247.2 dBc/Hz at TT process corner with the temperature of 27 °C with the voltage of 1 V from the proposed charge pump design. Due to the effects of current mode signalling, bias current regulation and differential signalling in the MCML circuits, it maintains a consistent output noise and phase noise spectrum, irrespective of changes in temperature, process corner and supply voltage.

### 4.4 Parametric analysis at different temperatures

The parametric analysis with different temperatures and voltage values for the proposed charge pump are analyzed using the Cadence Virtuoso tool at 45 nm CMOS technology. The performance of average power, propagation delay and noise values at different temperatures (− 90 °C, − 40 °C, 0 °C, 27 °C, and 90 °C) and supply voltages (0.8 V, 1 V and 1.2 V) of the proposed charge pump are tabulated in Table 4. The parametric analysis with different temperatures at different supply voltages are shown in Fig. 10.

### 4.5 Effect of process, voltage and temperature (PVT) on the proposed charge pump

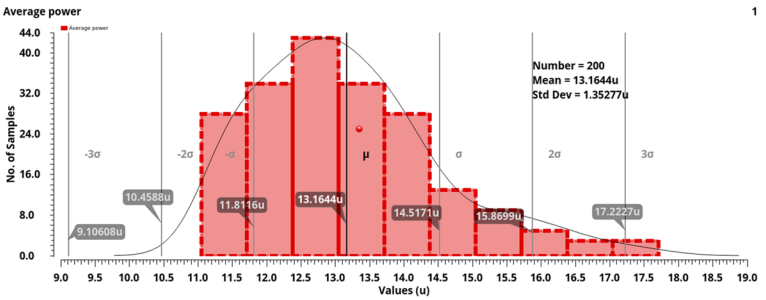
The performance of average power, propagation delay and noise values at different temperatures (− 40 °C, 27 °C and 90 °C), supply voltages (0.8 V, 1 V and 1.2 V) and process corners (TT, FF, SS) of the proposed charge pump are tabulated in Table 5. Figure 11 shows the analysis with different process corners at different temperatures and Fig. 12 illustrates the analysis with different supply voltages at different process corners.

### 4.6 Overall performance comparison

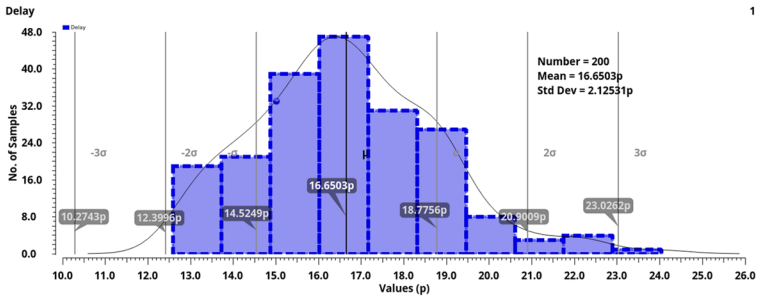
Table 6 represents the performance improvement of the proposed charge pump over the existing in percentage. The comparison result in Table 7 shows actual simulation results from the literatures. The layout representation of the proposed charge pump is depicted in Fig. 13 and the area consumption is 11.36 $\mu m^2$. The pre layout and post layout comparison of the proposed charge pump is illustrated in Fig. 14a, b for power and delay and Fig. 15a, b for output noise and phase noise respectively.



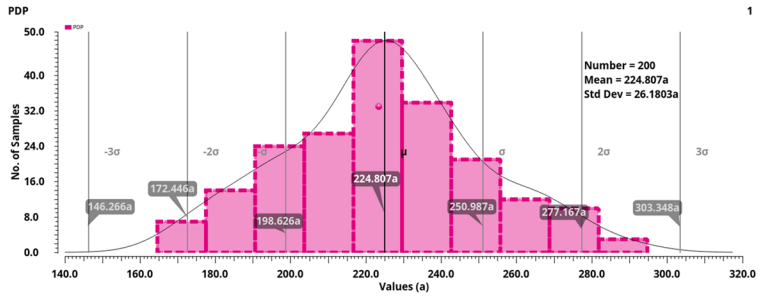
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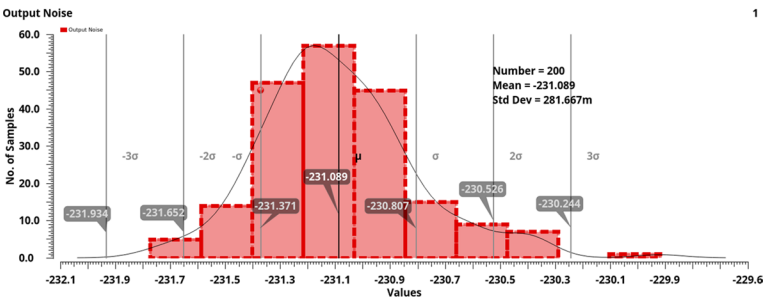

(a)



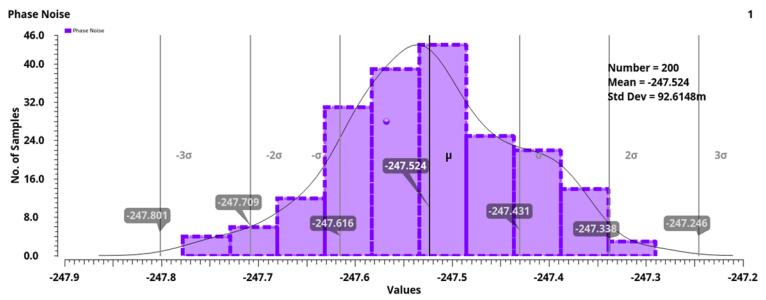

(b)




(c)




(d)




(e)

**Fig. 8** Monte carlo analysis of the proposed charge pump **a** Average power **b** Delay **c** PDP **d** Output noise **e** Phase noise



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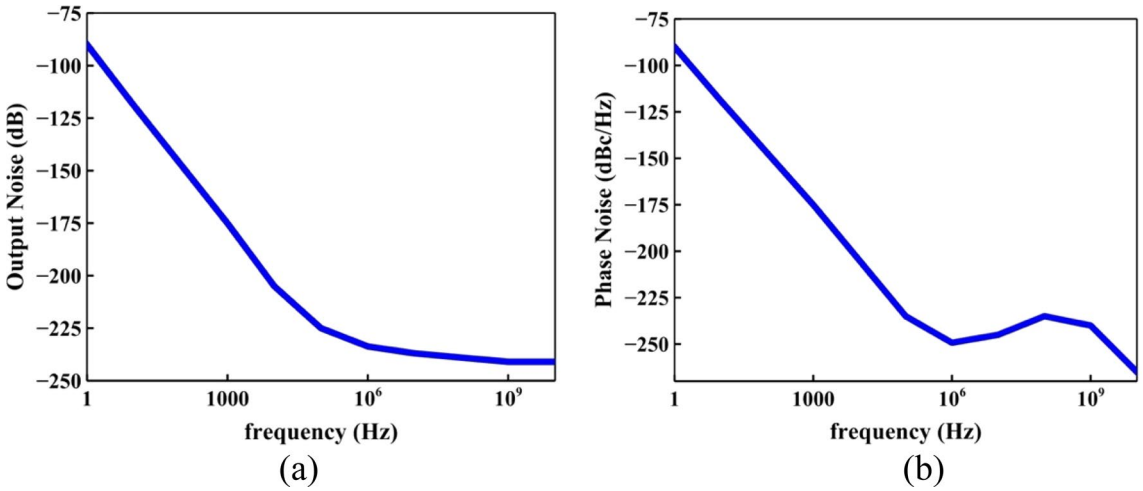



**Fig. 9** Noise analysis of the proposed charge pump **a** Output noise **b** Phase noise

**Table 4** Performance of power, delay and noise at different temperatures @ TT corner

| Vdd (V) | Temperature (°C) | Different parameters of the Proposed charge pump @ different temperatures | | | | |
|---|---|---|---|---|---|---|
| | | Power (μW) | Propagation delay (ps) | Power delay product (fJ) | Output noise (dB) | Phase noise (dBc/Hz) |
| *Pre layout* | | | | | | |
| 0.8 | − 90 | 8.59 | 15.98 | 0.137 | − 231.05 | − 248.3 |
| | − 40 | 8.1 | 21.95 | 0.177 | − 228.48 | − 248.1 |
| | 0 | 6.91 | 28.75 | 0.198 | − 228.29 | − 248.2 |
| | 27 | 5.25 | 39.24 | 0.245 | − 228.29 | − 248.2 |
| | 90 | 6.25 | 40.36 | 0.21 | − 227.5 | − 248.2 |
| 1 | − 90 | 15.27 | 5.37 | 0.081 | − 232.82 | − 247.9 |
| | − 40 | 14.84 | 7.26 | 0.107 | − 228.48 | − 247.7 |
| | 0 | 14.02 | 10.39 | 0.145 | − 228.29 | − 247.5 |
| | 27 | 13.19 | 16.71 | 0.221 | − 232.7 | − 247.2 |
| | 90 | 13.49 | 24.6 | 0.331 | − 227.5 | − 246.9 |
| 1.2 | − 90 | 25.9 | 3.17 | 0.082 | − 241.09 | − 247.6 |
| | − 40 | 25.64 | 6.79 | 0.174 | − 239.97 | − 247.3 |
| | 0 | 24.94 | 10.14 | 0.252 | − 238.71 | − 248.9 |
| | 27 | 23.67 | 12.66 | 0.307 | − 237.62 | − 248.5 |
| | 90 | 24.32 | 19.13 | 0.452 | − 234.21 | − 247.5 |
| *Post layout* | | | | | | |
| 0.8 | − 90 | 19.23 | 37.23 | 0.715 | − 227.94 | − 247.8 |
| | − 40 | 18.56 | 45.79 | 0.849 | − 223.71 | − 247.3 |
| | 0 | 16.84 | 49.72 | 0.837 | − 223.45 | − 247.6 |
| | 27 | 15.42 | 56.21 | 0.866 | − 223.3 | − 247.6 |
| | 90 | 16.24 | 61.73 | 1.002 | − 221.67 | − 247.7 |
| 1 | − 90 | 39.43 | 29.54 | 1.164 | − 228.43 | − 246.9 |
| | − 40 | 38.65 | 31.87 | 1.231 | − 223.14 | − 246.72 |
| | 0 | 37.92 | 34.91 | 1.323 | − 222.78 | − 246.51 |
| | 27 | 36.27 | 39.23 | 1.422 | − 228.32 | − 246.3 |
| | 90 | 36.81 | 45.9 | 1.689 | − 221.65 | − 245.76 |
| 1.2 | − 90 | 50.31 | 23.91 | 1.202 | − 237.63 | − 246.3 |
| | − 40 | 48.83 | 26.89 | 1.313 | − 236.21 | − 245.91 |
| | 0 | 47.21 | 29.21 | 1.379 | − 234.85 | − 246.4 |
| | 27 | 45.32 | 34.71 | 1.573 | − 234.12 | − 245.8 |
| | 90 | 46.91 | 39.23 | 1.84 | − 231.23 | − 245.3 |



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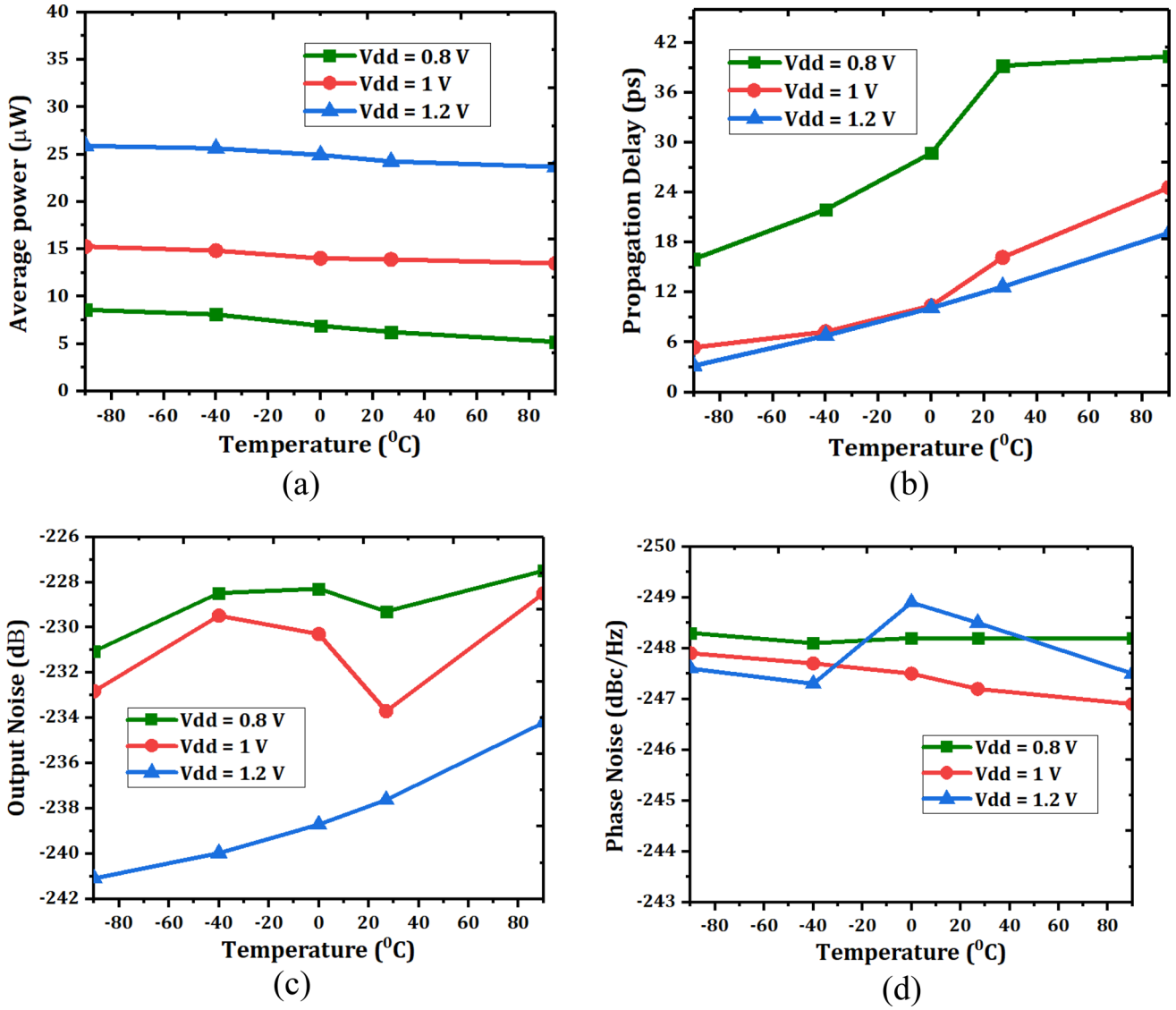



**Fig. 10** Parametric analysis at different temperatures of the proposed charge pump with **a** Average power **b** Propagation delay **c** Output noise **d** Phase noise



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**Table 5** Performance of proposed charge pump at PVT variation @ different corners

| Vdd (V) | Different parameters | Different parameters of the Proposed charge pump @ different temperatures @ different process corners | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | @ − 40 °C | | | @ 27 °C | | | @ 90 °C | | |
| | | TT | FF | SS | TT | FF | SS | TT | FF | SS |
| *Pre layout* | | | | | | | | | | |
| 0.8 | Power (µW) | 8.11 | 9.24 | 2.94 | 6.26 | 9.01 | 2.76 | 5.23 | 8.62 | 2.6 |
| | Delay (ps) | 21.95 | 12.69 | 23.97 | 26.97 | 15.32 | 29.2 | 31.3 | 17.9 | 33.5 |
| | PDP (fJ) | 0.178 | 0.117 | 0.07 | 0.168 | 0.138 | 0.08 | 0.163 | 0.154 | 0.087 |
| | Output noise (dB) | − 228.2 | − 232.9 | − 230.8 | − 228.3 | − 230.9 | − 226.7 | − 227.2 | − 227.2 | − 224.9 |
| | Phase noise (dBc/Hz) | − 247.8 | − 244.6 | − 248.2 | − 246.2 | − 244.1 | − 242.6 | − 245.4 | − 241.5 | − 247.9 |
| 1 | Power (µW) | 14.85 | 18.72 | 6.88 | 13.19 | 18.44 | 6.01 | 13.5 | 17.9 | 5.6 |
| | Delay (ps) | 6.93 | 2.27 | 12.12 | 16.71 | 11.54 | 21.39 | 24.4 | 19.6 | 29.47 |
| | PDP (fJ) | 0.102 | 0.042 | 0.083 | 0.221 | 0.212 | 0.128 | 0.329 | 0.35 | 0.165 |
| | Output noise (dB) | − 234.2 | − 234.6 | − 233.9 | − 232.7 | − 234.1 | − 232.4 | − 231.4 | − 228.6 | − 228.3 |
| | Phase noise (dBc/Hz) | − 248.3 | − 245.7 | − 249.1 | − 247.2 | − 245.3 | − 243.7 | − 246.5 | − 243.6 | − 249.6 |
| 1.2 | Power (µW) | 29.3 | 31.5 | 20.6 | 28.61 | 30.71 | 20.12 | 28.7 | 29.62 | 19.79 |
| | Delay (ps) | 3.86 | 1.37 | 5.28 | 11.26 | 9.24 | 12.92 | 19.36 | 13.4 | 24.59 |
| | PDP (fJ) | 0.113 | 0.043 | 0.108 | 0.322 | 0.283 | 0.259 | 0.555 | 0.396 | 0.486 |
| | Output noise (dB) | − 240.7 | − 241.1 | − 234.8 | − 234.7 | − 235.3 | − 235.1 | − 229.2 | − 229.6 | − 229.5 |
| | Phase noise (dBc/Hz) | − 250.2 | − 247.6 | − 251.2 | − 245.6 | − 243.4 | − 243.6 | − 249.3 | − 245.2 | − 251.7 |
| *Post layout* | | | | | | | | | | |
| 0.8 | Power (µW) | 18.56 | 21.32 | 9.78 | 15.42 | 19.51 | 9.23 | 16.24 | 18.82 | 8.73 |
| | Delay (ps) | 45.79 | 34.61 | 56.91 | 56.21 | 44.82 | 65.39 | 61.73 | 52.81 | 70.93 |
| | PDP (fJ) | 0.849 | 0.737 | 0.556 | 0.866 | 0.874 | 0.603 | 1.002 | 0.993 | 0.619 |
| | Output noise (dB) | − 223.7 | − 224.9 | − 221.4 | − 223.3 | − 225.6 | − 222.5 | − 221.6 | − 221.9 | − 220.3 |
| | Phase noise (dBc/Hz) | − 247.3 | − 243.8 | − 247.8 | − 245.6 | − 243.2 | − 241.9 | − 243.7 | − 240.2 | − 245.1 |
| 1 | Power (µW) | 38.65 | 42.91 | 17.85 | 36.27 | 42.73 | 17.32 | 36.81 | 39.71 | 16.25 |
| | Delay (ps) | 31.87 | 22.51 | 41.23 | 39.23 | 30.53 | 54.28 | 45.9 | 33.9 | 59.21 |
| | PDP (fJ) | 1.231 | 0.965 | 0.735 | 1.422 | 1.304 | 0.94 | 1.689 | 1.346 | 0.962 |
| | Output noise (dB) | − 223.1 | − 226.4 | − 220.6 | − 228.3 | − 231.7 | − 225.4 | − 221.6 | − 224.5 | − 220.1 |
| | Phase noise (dBc/Hz) | − 246.7 | − 245.4 | − 247.1 | − 246.3 | − 245.1 | − 243 | − 245.7 | − 242.2 | − 246.9 |
| 1.2 | Power (µW) | 48.83 | 52.67 | 23.6 | 45.32 | 49.51 | 22.81 | 46.91 | 50.32 | 21.93 |
| | Delay (ps) | 26.89 | 17.91 | 37.32 | 34.71 | 23.15 | 46.8 | 39.23 | 27.81 | 50.12 |
| | PDP (fJ) | 1.313 | 0.943 | 0.880 | 1.573 | 1.146 | 1.067 | 1.84 | 1.399 | 1.099 |
| | Output noise (dB) | − 236.2 | − 237.6 | − 234.8 | − 234.1 | − 236.9 | − 222.8 | − 231.2 | − 232.3 | − 230.9 |
| | Phase noise (dBc/Hz) | − 245.9 | − 244.7 | − 246.8 | − 245.8 | − 241.5 | − 241.8 | − 245.3 | − 241.5 | − 248.6 |



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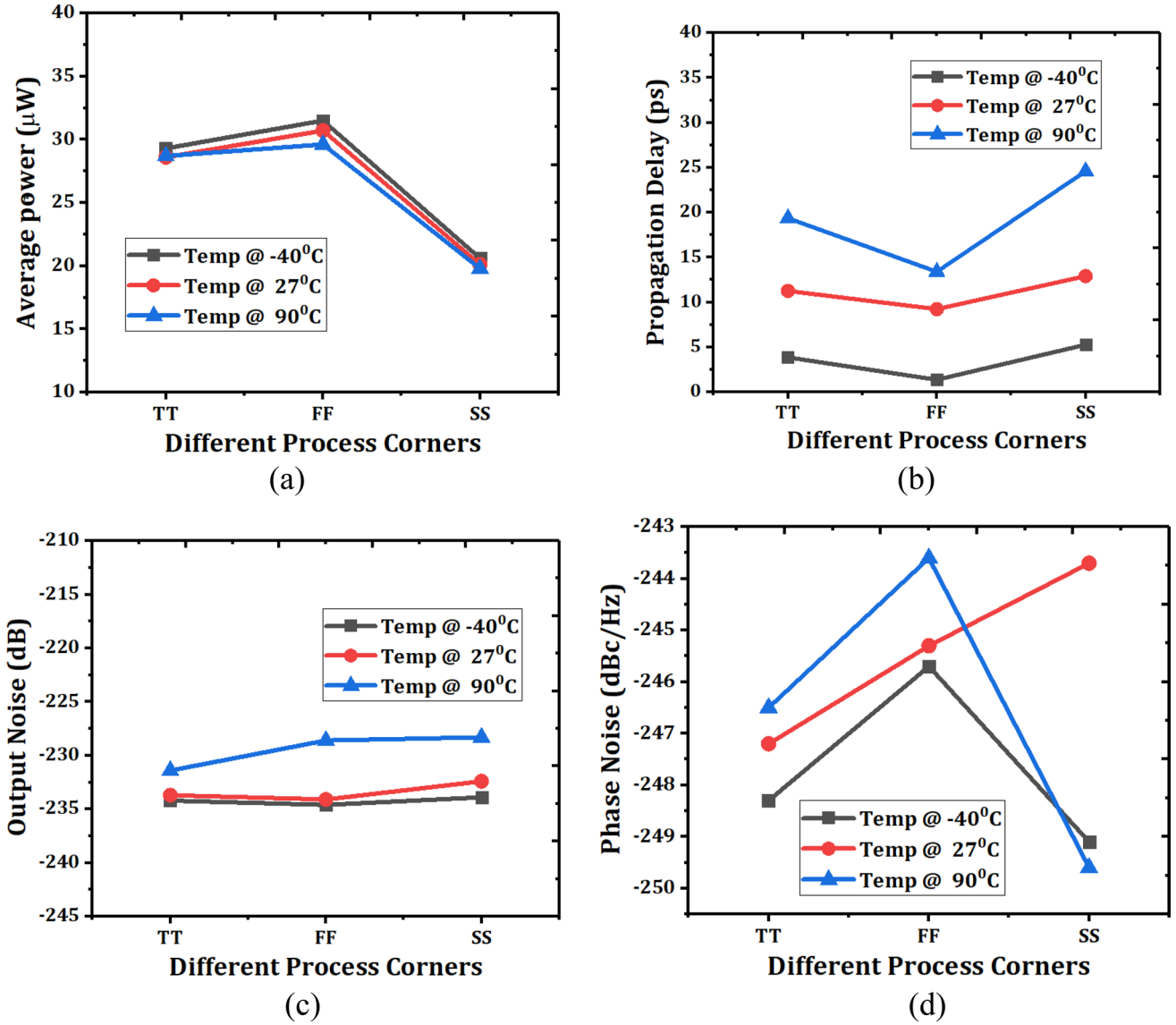



**Fig. 11** Performance of the proposed charge pump at different process corners with temperature on **a** Average power **b** Propagation delay **c** Output noise **d** Phase noise



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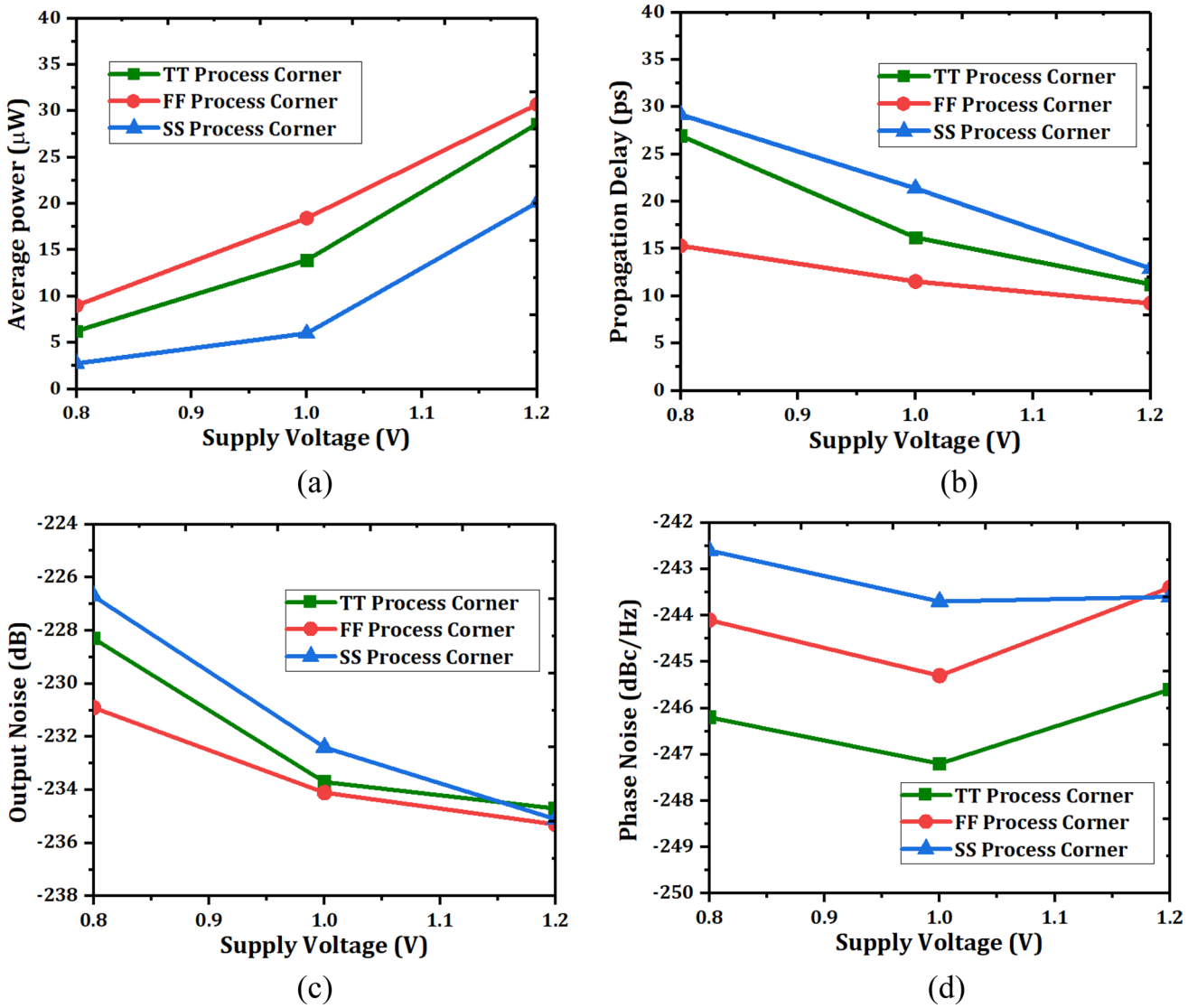



**Fig. 12** Performance of the proposed charge pump at different supply voltage with process corner on **a** Average Power **b** Delay **c** Output noise **d** Phase noise

**Table 6** Performance Improvement in Percentage of the proposed charge pump

| S. No | Parameters | Existing charge pump [18] | Proposed charge pump | Percentage improvement (%) |
|---|---|---|---|---|
| 1 | Power (μW) | 21.82 | 13.19 | 39.5 |
| 2 | Delay (ps) | 71.34 | 16.71 | 76.57 |
| 3 | PDP (fJ) | 1.56 | 0.221 | 85.8 |
| 4 | Output noise (dB) | − 214.8 | − 232.7 | 8.3 |
| 5 | Phase noise (dB) | − 227.6 | − 247.2 | 8.61 |

## 5 Conclusion

In this paper, a high-speed faster current driving charge pump is designed at 45 nm gpdk library with 1 V power supply. The proposed topology uses faster current driving approach with MCML logic to speed up the locking process. Simulations are performed using cadence virtuoso at 10 GHz of frequency. The proposed technique achieves 39.55% of power reduction, transient response speed improved by 76.57% and phase noise cancellation becomes 8.61% improvement over the existing circuit. It



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**Table 7** Comparison of proposed FCD-CP with existing simulation results

| Parameters | [21] | [22] | [24] | [20] | [19] | [25] | Proposed work |
|---|---|---|---|---|---|---|---|
| Technology (nm) | 130 | 90 | 90 | 180 | 90 | 90 | 45 |
| Supply voltage (V) | 0.5 | 1.2 | 1.2 | 1.8 | 1.8 | 1.2 | 1 |
| Output Swing (V) | 0.1–0.48 | – | 0.3–0.95 | – | 0–1.7 | 0.05–1.1 | 0–0.98 |
| Power (mW) | 0.06 | 0.134 | – | 22 | 1.74 | 0.095 | 0.013 |
| Phase noise (dBc/Hz) | – | – 90.214 | – | – 105.5 | – 101.0 | – 143.07 | – 247.2 |
| Frequency (GHz) | – | 1.25 | 2.5 | 2.4 | 2.5 | 6.5 | 10 |

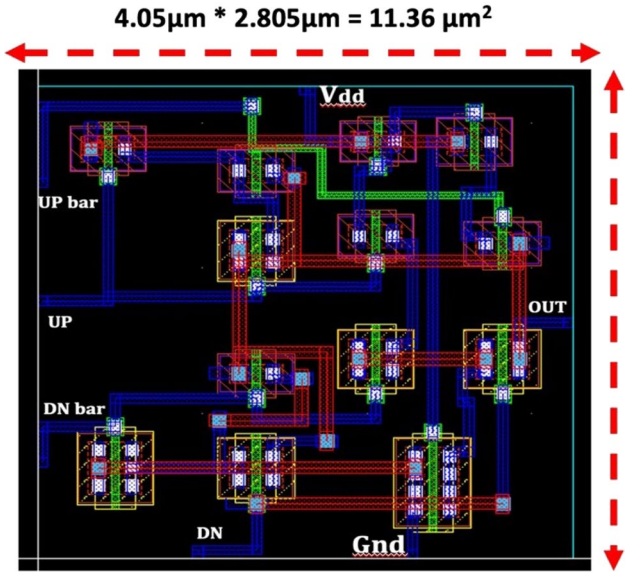



**Fig. 13** Layout design of the proposed charge pump

also achieves 98% of output swing which is more efficient compared to others. Simulation results are verified with the histogram plot of Monte-Carlo analysis and also validated with different corners along with different temperatures and voltages for the PVT tolerant analysis. The proposed method proved its efficient performance with the results achieved from the pre layout and post layout simulations using cadence virtuoso tool. Hence the proposed charge pump can be used in high frequency PLL application and also more suitable for millimetre wave frequency synthesizers.

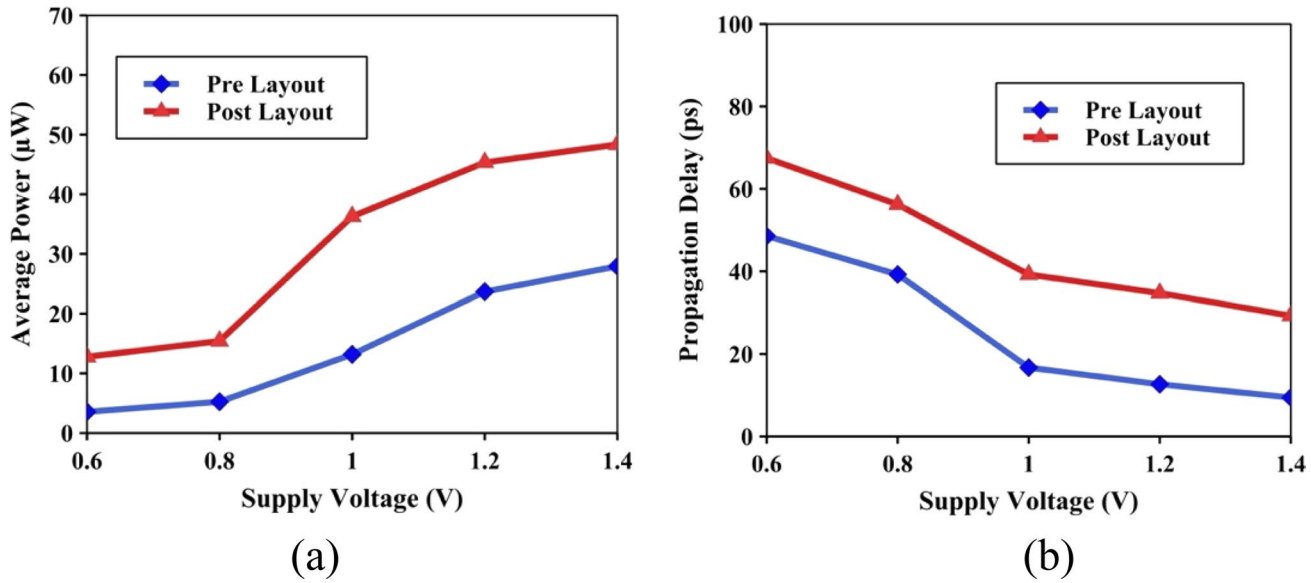



**Fig. 14** Pre-layout and Post-layout comparison of supply voltage Vs **a** Average power **b** Propagation delay



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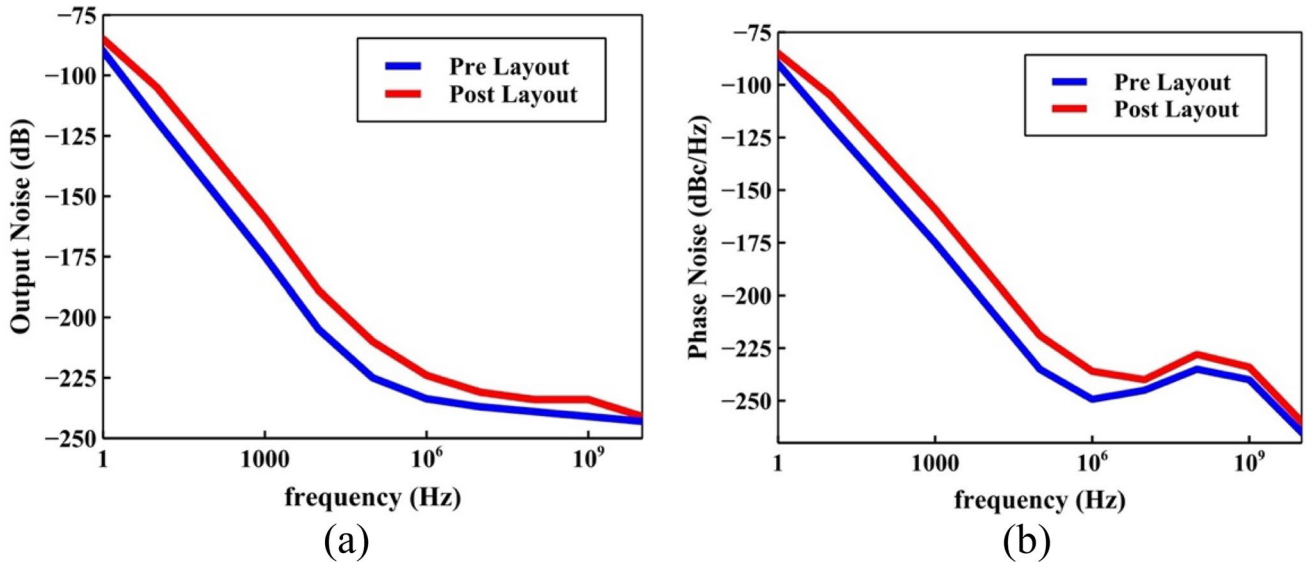



**Fig. 15** Pre-layout and Post-layout comparison of Frequency Vs **a** Output noise and **b** Phase noise

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**Data availability** The authors declare that the data related to this research is available in the manuscript.

## Declarations

**Conflict of interest** The authors declare that they have no conflict of interest.

**Ethics approval** The authors declare that all the procedures were followed in accordance with the ethical standards.

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